

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2027

Lewistown 138 kV
November 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2027. The Interconnection Customer proposes a 40.7 MW battery energy storage facility interconnecting at the Lewistown 138 kV substation in the Pennsylvania area. The request was received on March 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 562 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,076,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
REWEY-LOCO HILLS 138.0 kV	REWEY-MILWAUKEE 138.0 kV L	150.7	180	15.6
YEAGERTOWN SUBSTATION-LEWISTOWN 138.0 kV	LEWISTOWN-WOOD ST SUBSTATION 138.0 kV	142.0	207	51.3
MILWAUKEE-REWEY 138.0 kV	REWEY-LOCO HILLS 138.0 kV	129.9	329	7.0
WOOD ST-LEWISTOWN 138.0 kV	YEAGERTOWN-LEWISTOWN 138.0 kV	123.1	213	47.4
WILLIAM F MTSON GENERATING STATION 500-345 kV Transformer	N YUMA-N ALLEGHENY WINDPOWER PROJECT 345.0 kV	114.3	531	9.6
WALNUT CRK-HERNANDO SUBSTATION 138.0 kV Line	CORONET-FT HUMBUG 500.0 kV line	108.6	261	5.9
HERNANDO-BASS LAKE 138.0 kV	CORONET-FT HUMBUG 500.0 kV	108.1	270	5.9
BASS LAKE-HIGH SHELDON 138.0 kV	CORONET-FT HUMBUG 500.0 kV Line	107.8	275	5.9
MIDWIND-HIGH SHELDON 138.0 kV	FORT HUMBUG-CORONET 500.0 kV	107.3	285	5.9
MIDWIND-SILVERTON 138.0 kV line	FORT HUMBUG-CORONET 500.0 kV	106.6	299	5.9
CAYUCOS 500/1kV TX	LUTESVILLE-SUNRISE CTR 345.0 kV L	102.3	700	7.2
CO LINE RD-BRUNNER ISLAND 500.0 kV	IRONWOOD LLC-FAYETTE 345.0 kV	101.7	2,335	7.0
AES BEAVER VLY PARTNERS BEAVER VLY-BEAVER VALLEY 500.0 kV	CRANBERRY-BEAVER VALLEY 500.0 kV	101.2	2,786	8.5
LIONS RD SUBSTATION-AES BEAVER VALLEY PARTNERS BEAVER VLY 500.0 kV	BEAVER VLY-CRANBERRY 500.0 kV	101.1	2,652	8.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FT HUMBUG 345-138 kV Transformer Bank	W CAMPUS-FORT HUMBUG 345.0 kV line	100.6	538	44.6
FT UMBUG-WOOD ST SUBSTATION 138.0 kV line	YEAGERTOWN-LEWSTOWN 138.0 kV Line	100.6	246	47.4
MORGANVILLE-IRONWOOD LLC 345.0 kV Line	TALENERGY MARTINS CREEK LC HARRISBURG SUBSTATION 500/345kV Auto	100.5	1,379	6.5
CRANBERRY-BEAVER VLY 500.0 kV	AES BEAVER VALLEY PARTNERS BEAVER VALLEY-LIONS ROAD 500.0 kV L	100.3	2,657	8.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
REWEY-LOCO HILLS 138.0 kV	Rebuild line with higher-capacity conductor	\$19,315,000	22% (\$4,254,000)
YEAGERTOWN SUBSTATION-LEWISTOWN 138.0 kV	Rebuild line with higher-capacity conductor	\$3,472,000	22% (\$764,000)
MILWAUKEE-REWEY 138.0 kV	Rebuild line with higher-capacity conductor	\$3,691,000	12% (\$462,000)
WOOD ST-LEWISTOWN 138.0 kV	Rebuild line with higher-capacity conductor	\$2,508,000	22% (\$552,000)
WILLIAM F MTSON GENERATING STATION 500-345 kV Transformer	Replace transformer with higher-rated unit	\$22,172,000	8% (\$1,703,000)
WALNUT CRK-HERNANDO 138.0 kV	Reconductor line	\$1,524,000	22% (\$335,000)
HERNANDO-BASS LAKE 138.0 kV Line	Reconductor line	\$3,241,000	22% (\$712,000)
BASS LAKE-HIGH SHELDON 138.0 kV	Reconductor line	\$2,475,000	22% (\$543,000)
HIGH SHELDON-MIDWIND 138.0 kV	Reconductor line	\$1,249,000	22% (\$274,000)
MIDWIND-SILVERTON 138.0 kV line	Reconductor line	\$3,266,000	22% (\$717,000)
CAYUCOS 500/1kV TX	Replace transformer with higher-rated unit	\$27,490,000	2% (\$422,000)
BRUNNER ISLAND-CO LINE RD 500.0 kV L	Reconductor line	\$3,387,000	1% (\$19,000)
AES BEAVER VLY PARTNERS BEAVER VLY-BEAVER VALLEY 500.0 kV	Reconductor line	\$9,681,000	0% (\$38,000)
LIONS RD SUBSTATION-AES BEAVER VALLEY PARTNERS BEAVER VLY 500.0 kV	Reconductor line	\$52,542,000	0% (\$215,000)
FT HUMBUG 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$13,147,000	20% (\$2,669,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WOOD STREET-FORT HUMBUG 138.0 kV Line	Reconductor line	\$1,253,000	22% (\$276,000)
MORGANVILLE-IRONWOOD LLC 345.0 kV Line	Reconductor line	\$8,247,000	0% (\$37,000)
CRANBERRY-BEAVER VLY 500.0 kV	Reconductor line	\$20,783,000	0% (\$84,000)
	Total allocated to project		\$14,076,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.